Dec - 2025 CIE-II

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

Academic year 2025-2026 (Odd Semester)

Date	Dec 2025	Maximum Marks	50 + 10
Course Code	HS271TA	Duration	120 Min
Sem	VII	**CIE II**	

INDIAN KNOWLEDGE SYSTEM

PART – A (10 Marks)

Sl.No		M	CO	BT
1	**Which Doṣa is primarily responsible for movement in the body?** a) Pitta b) Kapha c) Vāta d) Tejas	1	3	L1
2	**The treatise that governs Vāstu Śāstra principles is:** a) Nāṭya Śāstra b) Mānasāra c) Sushruta Samhita d) Rigveda	1	3	L1
3	**The basic unit of Indian town planning in Vāstu is:** a) Maṇḍala b) Karmic grid c) Pada d) Prāṇa grid	1	3	L1
4	**The Eight Limbs of Vāstu are known as:** a) Aṣṭāṅga Vāstu b) Saddarśana c) Pañcha Mahābhūta d) Mantra Śāstra	1	2	L1
5	**The temple style with curvilinear Shikhara is:** a) Nagara b) Dravidian c) Vesara d) Ajanta	1	2	L1
6	**The ratio of circumference to diameter is known as:** a) e b) π c) φ d) α	1	3	L1
7	**Āryabhaṭa proposed that Earth:** a) Is stationary b) Rotates on its axis c) Orbits the Sun annually d) Is flat	1	2	L1
8	**Which metal was used first in ancient India?** a) Iron b) Copper c) Steel d) Aluminium	1	3	L1
9	**Which Yantra is used for measuring time?** a) Gola-yantra b) Chakra-yantra c) Water clock (Ghati Yantra) d) Shanku-yantra	1	3	L1
10	**The Chandasutras are fundamentally related to:** a) Geometry b) Meter of poetry c) Planetary motion d) Aromatic chemicals	1	2	L1

TEST

Sl. No.	Part- B Questions	M	CO	BT
1 a	Derive the nth term of an arithmetic progression	03	3	L1
b	Mention any two Panchanga elements.	02		
c	What is Wootz steel	02		
d	Explain Piṅgala's contribution to combinatorics	03		
2 a	Describe the role of geometry in temple construction.	03	3	L2
b	Describe the art of making natural perfumes.	03		
c	What is Piṅgala's binary system?	02		
d	What is a Tithi?	02		
3 a	Explain the features of Nagara and Dravidian temple architecture with examples.	03	2	L1
b	Describe major water management practices for sustainable agriculture	03		
c	Explain the BMI–Consciousness Complex with a suitable example	02		
d	Describe the Triguṇa System and its influence on human behaviour	02		
4 a	Compare Nagara and Dravidian temple architecture with examples.	03	2	L2
b	Describe the Triguṇa system and its psychological significance	02		
c	State components of the BMI-C model.	02		
d	Discuss the educational significance of the 64 Kalās.	03		
5 a	Mention any two of the 64 Kalās	02	3	L1
b	Differentiate between Tamas, Sattva and Rajas	03		
c	What is meant by Aṣṭāṅga Vāstu? Give two examples	03		
d	State two advantages of sprinkler irrigation.	02		

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	-	24	36	-	40	20	—	—	—	—

Course Outcomes:

CO1 Explain the nature, scope, and historical development of Indian Knowledge Systems and differentiate traditional knowledge from modern scientific paradigms

CO2 Analyze the foundational concepts of Indian contributions to linguistics, mathematics, and science, including works of Pānini, Piṅgala and others

CO3 Evaluate the applications of traditional Indian technologies in areas such as astronomy, metallurgy, architecture, and medicine.

CO4 Assess the relevance of Indian Knowledge Systems in the context of sustainable development goals and propose methods to safeguard and integrate IKS in contemporary society.

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DEPARTMENT OF INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 15th December 2025	**CIE – II**	**Max. Marks**	: 10 + 50
Semester : 7th	**UG**	**Duration**	: 30 + 90 Min
Course Title: Product Design and Development		**Course Code**	: IM372IA

Note:

1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	What is concept generation in product design?	01	L_1	CO3
2.	Name one traditional method used for concept generation.	01	L_1	CO3
3.	Name the two main methods of concept selection.	01	L_1	CO3
4.	Name the types of product specifications.	01	L_2	CO4
5.	What is the role of AI-assisted benchmarking in concept testing?	01	L_2	CO3
6.	What is the first step in concept testing? Selecting the target customer segment.	01	L_1	CO4
7.	Why is variety and supply chain consideration important in product architecture?	01	L_2	CO3
8.	What is product architecture?	01	L_2	CO4
9.	What are implications of product architecture?	01	L_3	CO3
10.	What is Generative AI-based concept ideation?	01	L_2	CO4
	Part – B			
1a.	You are designing an autonomous delivery drone. Establish three critical product specifications that represent a conflict among technical feasibility, user experience (UX), and legal constraints. Propose a single, weighted trade-off solution for one of the conflicts.	05	L4	CO1
b.	Critically evaluate the balance required between the need for measurable, quantitative specifications (metrics) and the consideration of subjective user experience (UX) quality when defining the product specifications for a new predictive maintenance industrial IoT sensor.	05	L_4	CO1
2a.	What are some different ways you could communicate a concept for a new user interface for an automotive audio system? What are the strengths and weaknesses of each approach?	06	L_3	CO4
b.	Roughly estimate N for the following product. List your assumptions. • A sleeping pillow for air travelers. Assume that there are 2000 jumbo jets in service and that most sleeping pillows will be used on jumbo jets.	04	L_4	CO3
3	What are the items to be included in the survey format to gather data from potential customers of a new product conceived by the design team? Why do you think respondents typically overestimate the likelihood that they will purchase a product? What advantages Survey has over the customer interviews for Concept Testing?	10	L_4	CO3
4a.	How can a design team use concept generation to develop a sustainable packaging solution for e-commerce?	05	L_4	CO4
b.	How does generative AI support concept ideation in the automotive industry?	05	L_3	CO3
5a.	Evaluate the trade-off between modularity and integration in a complex electromechanical product (e.g., a smart vacuum cleaner). Under what specific market condition (e.g., low volume vs. high variety) would the integral architecture still be the superior choice?	06	L_4	CO4
b.	A design team needs to quickly select the best concept among five options. Justify the choice of using Concept Screening (vs. Scoring) first, not merely for resource saving, but as a critical tool for achieving team consensus and aligning expectations.	04	L_4	CO3

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Date : 15/12/2025	**CIE – II**	**Max. Marks : 10 + 50**
Semester : VII	**UG**	**Duration : 30 + 90 Min**
Course Title: Total Quality Management		**Course Code : IM373TA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	Name any two QC tools commonly used in the seven-step problem-solving approach.	2	1	CO2
2.	Why is the catch-ball process important in Hoshin Management?	2	1	CO3
3.	Differentiate between proactive improvement and preventive action.	2	2	CO2
4.	A team finds recurring defects in a product. Which 8D steps should they focus on immediately to contain the issue and begin analysis?	2	3	CO3
5.	A manager wants to identify relationships among complex causes of a problem. Which Management and Planning Tool should they use?	2	3	CO4
	Part – B			
1	"The case study, discussed by Shoji Shiba, by the Broken Pellicle QI Team at Analog Devices illustrates the use of several TQM standards: the 7 steps, the most commonly used three of the 7 QC tools, and a standard presentation format called a quality improvement story or QI story". Provide a summary of learning from this case study.	10	4	CO2
2	How do you deploy semantics to distinguish between two kinds of language: affective language and the language of reports? Elucidate.	10	3	CO2
3	List and explain the seven principles for activating teamwork in an organisation.	10	2	CO3
4	Illustrate and evaluate the Proactive, Reactive, and Control Phases in Hoshin Management.	10	2	CO3
5	Evaluate the three categories of learning that are part of TQM learning system.	10	2	CO3

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Date	:	16th Dec 2025	**CIE – I**	**Max. Marks** :	**10 + 50**
Semester	:	7th	**UG**	**Duration** :	**30 + 90 Min**
Course Title: Predictive Analytics				**Course Code** :	**IM374TFA**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	What does accuracy measure in a classification model?	01	L_1	CO1
2.	What does precision measure in a classification model?	01	L_1	CO1
3.	In linear regression, what does the slope coefficient represent?	01	L_2	CO1
4.	Which metric is crucial in quality inspection models where missing defects is costly?	01	L_2	CO1
5.	Predicting employee attrition in HR analytics is an example of which of the six tasks od predictive analtyics?	01	L_2	CO1
6.	Which distance metric is commonly used in k-means clustering?	01	L_1	CO2
7.	What are the two main types of hierarchical clustering?	01	L_1	CO1
8.	Which clustering method does not require specifying the number of clusters in advance?	01	L_1	CO1
9.	What does "confidence" measure in association rules?	01	L_2	CO2
10.	What is the main limitation of Apriori algorithms?	01	L_2	CO2
	Part – B			
1 a.	State the primary objective function that the k-means algorithm attempts to minimize during the clustering process.	04	L_3	CO3
b.	Define lift as applied to association rule mining. In the context of market basket analysis, an analyst finds the rule Diapers→Beer has a high support and confidence, but a lift value close to 1.05. Explain what this lift value specifically indicates about the association between Diapers and Beer purchases.	06	L_3	CO3
2.	Write down the Mathematical form of the Sigmoid function and its graphical representation in the context of a Logistics Regression. Logistic Regression – Loan Approval Dataset: Income | Approved 30 | 0 50 | 1 70 | 1 Goal: Predict approval probability for Income = 60.	10	L_4	CO1
3.	What is the Significance of K in K-Means Algorithm? K-means with three clusters in 2D Data points: (X={(0,0),(0,3),(3,0),(3,3),(6,6)}) Initial centroids: (μ_1=(0,0), μ_2=(3,3), μ_3=(6,6)) Task: One assignment step; list clusters.	10	L_3	CO1

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4 a. What is the purpose of constructing a dendrogram to represent final clusters in Hierarchical Clustering?
Hierarchical clustering (single linkage) in 1D
Data points: (x={1,2,4,10})
Distance metric: Euclidean distance
Linkage: Single
Task: Build the dendrogram order and report clusters at threshold (t=3).

5. Consider a small grocery store transaction database. Each transaction includes items bought together. Let's analyze the following dataset:

Transaction ID	Items Bought
1	Milk, Bread, Butter
2	Milk, Bread
3	Bread, Butter
4	Milk, Butter
5	Bread, Butter, Jam

i. Calculate Support for all the items in the itemset above.
ii. Generate Frequent Itemset
iii. Generate Association Rules and Calculate the confidence of these Association Rules.
iv. Calculate Lift.

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Academic year 2025-2026 (ODD Sem)

DEPARTMENT OF INDUSTRIAL ENGINEERING & MANAGEMENT

Date	**16.12.2025**	**Maximum Marks**	**10 + 50**
Course Code	**IM375TGM**	**Duration**	**120 Mins**
Sem	**VII Semester**	**CIE – II**	
PROJECT MANAGEMENT (Institutional Elective)			

Note:

1. Answer all the Questions.

PART – A

Sl. No.	Questions	M	BT	CO
1	Why is a project life cycle important?	02	L1	CO1
2	What is the primary output of the Plan Quality Management process?	02	L2	CO1
3	Mention the key function of the Change Control Board (CCB).	02	L2	CO2
4	List two organizational factors that influence project management.	02	L1	CO1
5	What is the difference between corrective action and preventive action?	02	L2	CO2

PART – B

Sl. No.	Questions	M	BT	CO
1.	During execution of a highway construction project, the project faces delay due to land acquisition issues. Explain how stakeholder management and governance frameworks help resolve such critical external dependencies.	10	L3	CO3
2.	Explain how different organizational structures (Functional, Matrix, Projectized) influence the decision-making speed and resource allocation of a project. Apply the concepts to a real-world example.	10	L2	CO2
3.	A consumer electronics company wants to launch a new Smart Home Security Device (AI-based door camera + alarm system). As the Project Manager, prepare a detailed Project Charter for this new product development project. Include all key components of a formal project charter and explain how each element is applied to this situation.	20	L4	CO4
4.	A company is developing an online Fee Payment Portal for a university. As the Project Manager, explain how you will apply the three processes of Project Quality Management — Plan Quality Management, Perform Quality Assurance, and Control Quality — to ensure high-quality software delivery. Provide suitable software-related examples for each stage.	10	L4	CO4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Max Marks	Quiz	06	04	--	--	--	04	06	--	--	--	--
		Test		10	10	30	--		10	10	30	--	--

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Date : 16 Dec 2025	**CIE – II**	**Max. Marks : 10 + 50**	
Semester : VII	**UG**	**Duration : 30 + 90 Min**	
Course Title: Principles & Practices of Cyber Law		**Course Code : HS375TGT**	

Note: Answer all questions

Q.No.	PART- A	M	BT	CO
1	Define Digital Signature.	2	2	CO4
2	Shreya signed an online contract for a product, but later she wanted to dispute the contract terms since the contract was made electronically. Can the contract be challenged under the IT Act, 2000?	2	3	CO4
3	Give an example of technological advancement that led to the use of Electronic Signatures.	2	2	CO3
4	A suspect denies being present at a crime scene, but investigators suspect he used Google Maps that night. What mobile evidence can help verify his location?	2	3	CO3
5	What is the role of Interpol in cybercrime investigation?	2	2	CO4
	PART -B			
1	Explain the scope and importance of Digital Forensics in investigating cybercrimes.	10	02	CO3
2	A company discovers that sensitive financial files have been leaked. The IT team suspects that an employee copied the files to a USB drive before resigning. Investigators seize the employee's workstation and USB drives for analysis. a. Which type of forensic analysis should be conducted on the workstation? b. How can investigators determine if files were copied to the USB drive? c. Which tools can be used to acquire and examine digital evidence?	10	04	CO3
3	Discuss the importance of FIR and Charge Sheet in cybercrime prosecution .Explain the procedure for filing an FIR in cybercrime cases.	10	03	CO3
4	Discuss the relevance of signature in the electronic era, highlighting the difference between handwritten and digital signatures, and the technological advancements that have influenced their development.	10	03	CO2
5	A customer, Priya, purchases a laptop from an e-commerce website "TechBuy.in." She makes an online payment through a digital wallet, and the website issues an electronic invoice. However, after delivery, Priya discovers that the laptop is defective and that the return request she submitted through the website is ignored. When she contacts customer service, she receives no response. Priya files a complaint claiming that the e-commerce company failed in its legal duty under the *Information Technology Act, 2000* and violated her consumer rights. Based on this case, answer the following: **(i)** Explain how the *Information Technology Act, 2000* provides a legal framework for E-Commerce And Electronic Transactions. **(ii)** Discuss the key issues and challenges in regulating e-commerce in India as highlighted by this case. **(iii)** Suggest legal and practical measures that can protect consumers like Priya in the digital marketplace.	10	04	CO3

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Date : 16-12-2025	**CIE – II**	**Max. Marks**	**: 10 + 50**
Semester : VII	**UG**	**Duration**	**: 30 + 90 Min**
Course Title: Sustainability Engineering & Circular Economy (Elective)		**Course Code**	**: IM374TFD**

Note:
1. Answer all the Questions.

Sl. No.	Questions	M	BT	CO
	PART – A			
1	What is a conceptual standard framework?	2	2	3
2	What is sustainable supply chain management?	2	2	3
3	List two impacts of climate change on engineering.	2	1	3
4	What is the 3R model in the context of circular economy?	2	2	4
5	What is meant by eco-design?	2	2	4
	PART – B			
1.	What are the key steps involved in formulating sustainability standards? How do international and national bodies collaborate to ensure these standards are credible, applicable, and effective across industries? Illustrate your answer with examples of standard-setting frameworks or organizations.	10	2	3
2.	Why is corporate sustainability becoming a strategic priority for modern enterprises? How does integrating environmental, social, and governance (ESG) principles enhance long-term business performance and stakeholder trust? Discuss with relevant examples and frameworks.	10	3	3
3.	How do global mega-trends such as urbanization, climate change, digitalization, and demographic shifts shape the future of sustainable engineering? In what ways must engineers adapt their design, operations, and decision-making to align with these evolving trends? Discuss with relevant examples and implications.	10	4	3
4.	How do smart technologies such as IoT, AI, and blockchain contribute to the implementation of circular economy principles? In what ways can these technologies support resource optimization, waste reduction, and closed-loop systems in engineering and industrial contexts?	10	4	4
5.	What is a digital factory, and how does it transform traditional manufacturing systems? In what ways does the integration of digital technologies in factory operations contribute to sustainability?	10	3	4

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DEPARTMENT OF

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Date : 16th Dec 2025	**CIE-II**	**Max. Marks : 10 + 50**	
Semester : VII	**UG**	**Duration : 30 + 90 Min**	
Course Title: Artificial Intelligence in Industrial and Systems Engineering		**Course Code : IM374TFB**	

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1	Traditional optimization techniques often rely on fixed mathematical models, whereas AI-based optimization uses _____ to learn pattern	1	3	01
2	AI-driven demand forecasting reduces forecasting error by using large-scale _____ data	1	2	01
3	Reinforcement learning is often used in production scheduling to maximize cumulative _____.	1	2	02
4	AI-based inventory management helps determine safety stock using real-time ___data.	1	3	02
5	In logistics optimization, route planning can be enhanced through AI using __ algorithms.	1	2	02
6	In automotive quality control, ___ vision is widely used for surface defect detection.	1	3	01
7	Aerospace condition monitoring makes use of ____ frequency analysis to detect engine issues.	1	2	01
8	Time-series forecasting in predictive maintenance frequently uses _____ models.	1	2	02
9	Image-based defect detection systems require accurate _____ to mark true defects.	1	2	03
10	Feature extraction in anomaly detection transforms raw signals into meaningful _____.	1	2	02
	Part – B			
1	Explain AI-powered approaches for demand forecasting and inventory management.	10	3	01
2	Discuss AI-based production scheduling and resource allocation techniques with examples.	10	3	02
3	Describe the limitations of traditional optimization techniques in supply chain management.	10	2	03
4	Explain the importance of sensor fusion in predictive maintenance.	10	3	03
5	Explain end-to-end predictive maintenance using ML: data, models, diagnostics, forecasting, and decision-making.	10	2	03

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Department of Electrical & Electronics Engineering

Date: 16/12/2025	**Test - 2**	**Max. Marks:50+10**
Semester:7	**UG**	**Duration: 1 H:50 M**
Course Title: e-Mobility		**Course Code:EE375TGH**

SL No.	PART-A QUIZ QUESTIONS	M	BT	CO
1	List the essential functions of BMS.	1	L1	CO2
2	A 100 Ah battery delivered 20 Ah. Calculate SOC.	1	L2	CO2
3	List any two electrochemical reactions involved in a Li-ion battery	1	L2	CO1
4	State any two requirements of energy storage in hybrid electric vehicles.	1	L1	CO1
5	What is the key advantage of supercapacitors over batteries?	1	L1	CO2
6	What is hybridization of energy storage systems? Give one example.	1	L1	CO3
7	Differentiate between SOC (State of Charge) and SOH (State of Health)	2	L1	CO3
8	Differentiate between battery-based and fuel-cell-based energy storage systems.	2	L2	CO2

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PART-B TEST QUESTION

		M
1. a	Explain the redox reaction in a battery? Explain the basic structure of a Battery.	6
b	What is the voltage range for the cell when used in a HEV application with a DOD of 25% to 75% and a load ranging from no-load to a full load of 6C? What is the battery pack voltage if there are 192 cells arranged with 96 cells in series and two strings in parallel?	4
2. a	Explain the following regarding the charging of Lithium-ion batteries (a) Influence of Charging voltage on charging process. (b) Influence of charge current on charging process.	4
b	Explain the different types of Battery Management System (BMS) topologies Centralized, Modular, and Distributed .along with their advantages and disadvantages.	6
3. a	With neat schematic explain the concept of Active balancing and passive balancing of batteries.	6
b	A 24 kWh battery pack can be fast charged from 0% to 80% SOC in 30 min. Determine the approximate charge current and power in order to achieve this charge time.	4
4. a	Draw and explain the block diagram of a complete SRM drive system including the motor, converter, controller, and feedback loops. Discuss how each block contributes to stable and efficient operation.	10
5. a	. With neat block diagram, explain the working of a Fuel Cell Energy Storage System for EV propulsion.	6
b	List any four performance parameters used to evaluate energy storage systems in EVs.	4

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DEPARTMENT OF MECHANICAL ENGINEERING AY 2025-26 (ODD SEM)				
Date	**16/12/2025 Friday 1.20 PM to 3.20 PM**	**CIE 2**	**Max. Marks**	**10+50**
Semester	**VII**	**UG**	**Duration**	**2 Hrs.**
Course	**INDUSTRY 5.0 (Institutional Elective)**		**Course Code**	**ME375TGP**

#	QUIZ	M	BT	CO
1	Which statement best describes a digital twin?	1	1	1
2	Mention a key feature that distinguishes a digital twin from a standard simulation.	1	1	1
3	Highlight a security challenge associated with digital twins.	1	2	2
4	The use of AI-driven supply chain optimization leads to which benefit?	1	2	2
5	What is a major ethical concern regarding the increased use of AI and automation in manufacturing?	1	2	2
6	What is edge computing?	1	1	1
7	How does a circular economy primarily address greenhouse gas emissions?	1	2	2
8	What is the term for a systematic approach that assesses a product's environmental impact throughout its entire lifespan, from raw material extraction to disposal?	1	1	1
9	What are the Sustainable Development Goals (SDGs)?	1	2	2
10	What is the primary focus of the Global Reporting Initiative (GRI)?	1	2	2

#	TEST	M	BT	CO
1	Explain the working of digital twins. Also discuss the Key applications and benefits across industries	10	2	3
2	What do you mean by Artificial Intelligence (AI) & Machine Learning (ML)? Explain the application of AI and ML in (i) Predictive maintenance (ii) Supply chain optimization (iii) Product development and innovation (iv) Human-robot collaboration	10	4	2
3	What are Cyber-Physical Systems? Discuss the core components of CPS and the applications of CPS	10	2	3
4	Briefly discuss the principles of (i) Sustainable and Resilient manufacturing (ii) Circular economy (iii) Life cycle assessment	10	3	2
5	Explain the regulatory frameworks and standards for sustainable practices with examples.	10	4	

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Date: 16-12-25	**CIE-II**	**Max. Marks: 10+50**
Semester: VII	**UG**	**Duration: 2 Hrs**
Course Title: Statistical Methods for Engineers		**Course Code: MA375TGO**

Department of Mathematics

Sl. No.	QUIZ	M	BT	CO
1	You are performing an ANOVA Analysis, and the total sum of squares is 36 and the treatment sum of squares is 16, what would the error sum of squares be?	1	2	1
2	In ANOVA analysis, the degrees of freedom associated with the numerator of the F-value are calculated using which formula?	1	1	1
3	The one-way ANOVA Analysis below has 10 treatment groups with the total degrees of freedom of 19. Variation Source | Sum of Squares (SS) | Degrees of freedom (DF) | Mean Squares (MS) | F - Value Treatment(Between) | | | | | Error (Within) | 55 | | | | Total | 100 | 19 | | | Calculate the Treatment Mean Square for this ANOVA Table.	2	2	4
4	In a 2^3 factorial experiment with three factors A, B and C, each at two levels (0 and 1), write the treatment combinations for: (i) A and C at the first level and B at the second level. (ii) A and C at the second level and B at the first level.	2	1	3
5	Classify the data point (2.5, 5.5), given the linear score functions for two populations as $\hat{S}_a^L(x) = -441.6006 + 156.3833\,x_1 + 63.3446\,x_2$ $\hat{S}_b^L(x) = -292.5690 + 127.5972\,x_1 + 51.3522x_2$.	2	1	1
6	Given the covariance matrix for a data with 3 variables X_1, X_2, X_3 as $\begin{bmatrix} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ and the first principal component as $\begin{bmatrix} 0.383 \\ -0.924 \\ 0 \end{bmatrix}$, then the correlation of X_1 with Y_1 and correlation of X_2 with Y_1 are ___ and ___ respectively. λ are 5.83, 2, 0.17	2	1	2

Sl. No.	TEST	M	BT	C
1	A local bank has three branch offices. The bank has a liberal sick leave policy, and a vice-president was concerned with employees taking advantage of this policy. She thought that the tendency to take advantage depended on the branch at which the employee worked. To see if there were differences in the time employees took for sick leave, she asked each branch manager to sample employees randomly and record the number of days of sick leave taken during 1990. Ten employees were chosen, and the data are listed below. Does this data indicate a difference in branches? Use a level of significance of 0.05. [Given $F_\alpha(df_{k-1}, df_{N-k}) = 4.74$] Branch 1 | 15 | 20 | 19 | 14 Branch 2 | 11 | 15 | 11 |	10	2	

A researcher wants to test four diets A, B, C, D on growth rate in mice. The mice are divided into 3 groups according to their weights. Heaviest 4 mice are put in Block I, Block II and Block III respectively. Within each block, diet is given at random to the animals. After 15 days, increase in weight is given in the following table:

Blocks	Treatments/Diets			
	A	B	C	D
I	12	8	6	5
II	15	12	9	6
III	14	10	8	5

Perform a two-way ANOVA to test whether the data indicate any significant difference between the four diets due to different blocks. Use a level of significance 1%.

Given: $F_{0.01}(2,6) = 10.925$ & $F_{0.01}(3,6) = 9.780$

An experiment was planned to study the effect of Sulphate, Potash and Superphosphate on the yield of peanut. All the combinations of 2 – levels of Superphosphate (0 cwt (p_0) and 1 cwt (p_1) per acre) and two levels of Sulphate and Potash (1 cwt (k_0) and 2 cwt (k_1) per acre) were studied in a randomized block design with 4 replications each. The yields in appropriate units obtained are given in the following:

Blocks	Yield (lbs per plot)			
	Treatment combinations			
	(1)	k_1	p_1	k_1p_1
I	23	25	27	32
II	26	31	29	34
III	19	21	24	30
IV	28	30	26	36

Analyse the data of a 2² factorial experiment conducted using a RBD and give your conclusions. Use a level of significance of 1%. Given $F_{0.01}(3,9) = 8.12$

Obtain the main effects and the interaction effect in the following two dimensional data:

2	4	2	6
1	6	3	8

Obtain the simple effects of each of the two factors from the data given in Q.1 above and hence compute the percentage contribution of each of the simple effects and interaction effect.

Department of Civil Engineering

Date: 16-12-2025	CIE- II	Max. Marks: 60
Semester: VII	UG	Duration: 2 Hrs
Course Title: Freight Transportation Systems and Logistics		Course Code: CV374TGH

Quiz

S No	Questions	M	BT	CO
1.	Explain the term freight capacity issues with an example.	2	2	2
2.	How does freight productivity influence overall logistics performance?	2	2	2
3.	Briefly describe the role of scheduling in freight transportation.	2	2	3
4.	List any two current trends in logistics and explain them.	2	2	3
5.	What is the aim of logistics in a supply chain context?	2	2	3

S No	Questions	M	BT	CO
1.	Explain the factors affecting freight productivity and performance. Illustrate how organizations measure these factors.	10	3	3
2.	Describe the process of route planning, scheduling, collection, storage, and distribution center operations in logistics networks.	10	3	2
3.	Explain strategic decisions in logistics and discuss the steps involved in designing an effective logistics strategy.	10	3	3
4.	Compare various competing modes of transport for specific commodities. How does mode selection affect cost and service levels?	10	3	2
5.	Evaluate the importance of distribution centers in improving freight performance. Provide examples of their strategic roles.	10	3	3

RV College of Engineering

Department of Electronics and Telecommunication Engineering

Date: 16/12/25	**CIE - 2**	**Max. Marks: 60**
Semester: 7	**UG**	**Duration: 2 Hrs**
Course Title: Space Technology and Applications		**Course Code: ET375TGL**

Q.No	Quiz-2	M	BT	CO
1	List the classification of satellites based on service.	2	L2	CO2
2	Mention the difference between linear bent pipe transponder and baseband processing transponder.	2	L2	CO2
3	What is the main function of command systems in satellites?	2	L2	CO2
4	What is multiple access? Mention its advantages.	2	L2	CO3
5	What is a transponder?	2	L2	CO3
	Test-2			
1	Illustrate the altitude and orbit control subsystem present in the satellite with a neat diagram.	10	L3	CO2
2	With a neat diagram, explain the linear bent pipe transponder.	10	L2	CO2
3a	Explain the functions of tracking systems in the satellite. List and explain the different types of tracking methods.	6	L2	CO2
3b	What are the main functions of telemetry and monitoring systems?. List the various data collected to check the status of the satellite.	4	L2	CO2
4	Explain the different types of multiple access techniques used for satellite communication systems.	10	L2	CO3
5	With a neat block diagram, explain the DBS receiver system.	10	L2	CO3

BT- Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	-	36	24	-	-	50	10	-	-	-

RV College of Engineering

Department of
Aerospace Engineering

Date	14th December 2025	Maximum Marks	60
Course Code	AS375TGB	Duration	120 Min
Course Title	AIRCRAFT SYSTEMS	Semester	VII
ACY	2025-26	Quiz &Test-II	

PART -A

Sl No	Questions	M	BT	CO
1	What is the primary purpose of an APU in an aircraft during ground operations?	2	2	3
2	Differentiate between electric and pneumatic starting systems.	2	2	2
3	Why is it important to achieve a minimum rpm before introducing fuel during start?	2	3	1
4	Name any two major components of an aircraft fuel system and state their functions.	2	3	3
5	What triggers the deployment of a RAT in modern aircraft?	2	2	4

PART-B

Sl.No.	Questions	M	BT	CO
1	Describe the working principle of Pressure Feed Fuel Systems with required sketch.	10	L2	2
2	Explain Briefly about the Magneto – Ignition System & FADEC system	10	L1	1
3	With a neat sketch explain wet sump lubrication system	10	L2	3
4	Explain Pressure measurements and indicating systems with a neat sketch	10	L3	3
5	Explain Mach Warning system and altitude alerting System	10	L3	4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	12	12	24	12	10	38	24	0	0	

RV College of Engineering

Department of Civil Engineering

Date	16.12.2025	CIE – II (Test + Quiz)	Maximum Marks	50+10
Semester	VII	UG (Institutional Elective)	Duration	90+20 Min
Solid Waste Management and Statutory Rules			Course Code : CV375TGG	

Note: Answer all questions. Assume missing data if any suitably.

	PART A			
Q.No.	Questions	M	BT	CO
1.	State the characteristics of Hazardous waste.	2	2	1
2.	Classify the methods of landfilling.	2	3	1
3.	Compare aerobic and anaerobic composting.	2	3	4
4.	Define Leachate and Hazardous waste.	2	1	3
5.	List any 4 factors influencing anaerobic composting process.	2	2	4

	PART B			
Q No.	Questions	M	BT	CO
1.a	List and explain classification of Hazardous Waste with examples.	5	2	3
1.b	With the help of neat sketch enumerate the characteristics of onsite Hazardous Waste storage bin.	5	2	3
2	Enumerate salient features of Hazardous and Other Wastes (Management and Transboundary Movement) Rules, 2016 rules	10	2	1
3	With the help of screening model illustrate identification of HW. Explain terminologies.	10	3	3
4	With the help of neat sketch explain control of gas movement by permeable methods.	10	3	4
5a	Determine the amount of oxygen required to compost 1000kg of organic solid waste aerobically. Assume that initial composition of the organic material to be decomposed is given by $[C_6H_7O_2(OH)_3]_5$ and the final composition is expected to $[C_6H_7O_2(OH)_3]_2$ and that 400Kg of material remains after composting process. Prove the same with proper check.	5	3	3
5b	List and briefly explain the factors to be considered in evaluating potential landfill sites.	5	2	1

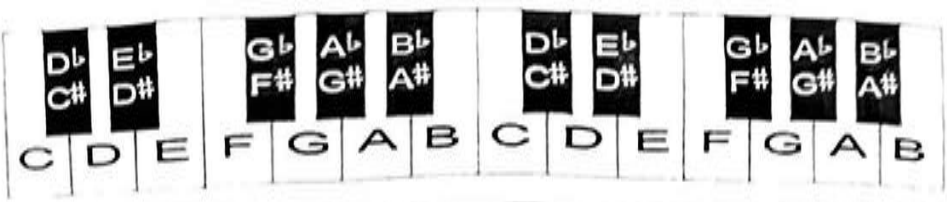
BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	12	5	27	16	02	29	29	-	-	-

RV College of Engineering

Date: 16-12-2025	CIE 2	Max. Marks: 10+50
Semester: VII	UG	Duration: 2 Hrs (1:20PM- 3:20PM)
Course Title: Mathematics of Music		Course code: MA375TGR

Department of Mathematics



No	Quiz	M	BT	CO
1.	Construct the swara sequence for the 52nd Melakartha (Ramapriya).	2	1	2
2.	Write the continued fraction expansion of $log_2(3/2)$ up to three terms.	2	2	1
3.	Give the Fourier series form of a periodic function and explain what the coefficients represent physically.	2	2	2
4.	Define the spectral centroid and compute it for a harmonic spectrum with amplitudes $A = \{5,3,1\}$ at frequencies $\{f, 2f, 3f\}$.	2	3	3
5.	Define a pitch class and express pitch class addition formally in modulo 12.	2	2	3

S No	Test	M	BT	CO
1a	Prove that $log_2(3/2)$ is irrational using proof by contradiction. Explain in detail the musical consequence of it.	10	2	1
2a	Derive the formula for (or explanation on) the total number of Melakartha rāgas (72) starting from the combinatorial possibilities of swaras, and elaborate on the *Katapayadi sutra* therein.	10	3	2
3	Derive the mathematical explanation for difference tones using the nonlinear distortion equation: $y = ax + bx^2$. If two pure tones at frequencies f_1 and f_2 are input, show that the output contains the difference frequency $\lvert f_1 - f_2 \rvert$. When two notes at 600 Hz and 800 Hz are played simultaneously: (i) Calculate the first-order difference tone. (ii) Calculate the first-order summation tone. (iii) Identify the musical interval represented by 600 Hz and 800 Hz. Also, explain why combination tones are more prominent in organs and brass instruments.	10	3	4

4	For a square wave $f(t) = \begin{cases} k \text{ for } 0 < t < \frac{T}{2} \\ -k \text{ for } \frac{T}{2} < t < T \end{cases}$, derive its Fourier series representation showing that it contains only odd harmonics with amplitudes decreasing as $1/n$. Explain why the square wave sounds different from a sawtooth wave despite both being periodic.
5	Prove that the set of all transposition operations $\{T_0, T_1, T_2, \ldots, T_{11}\}$ forms a cyclic group under composition, isomorphic to $(\mathbb{Z}_{12}, +)$. Show that: (i) Closure: $T_m \circ T_n = T_{(m+n)} \bmod 12$ (ii) Associativity holds (iii) Identity element exists (iv) Inverse elements exist

Academic year 2025-2026(ODDSem)

DEPARTMENT OF

ELECTRONICS AND INSTRUMENTATION ENGINEERING

Date	16.12.2025	Maximum Marks	10+50
Course Code	EI375TGK	Duration	120 Min
Sem	VII	CIE (Quiz &Test) - 2	

Course Title: **DISEASE AND DIAGNOSTICS AN ENGINEERING PERSPECTIVE**

Sl. No.	PART-A	M	BT	CO
1.1	What is spirometry?	1	L1	2
1.2	Define Tidal Volume (TV) and Residual Volume (RV).	2	L1	1
1.3	What is Functional Residual Capacity (FRC)? Mention its components.	2	L2	1
1.4	Which lung capacity cannot be measured using spirometry?	1	L1	1
1.5	Name two types of dialysis.	1	L1	2
1.6	Define kidney filtration.	1	L1	3
1.7	What is a dialyzer?	1	L1	3
1.8	What is creatinine?	1	L1	3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	05	02	03	-	8	2	-	-	-	-

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Sl. No.	PART- B	M	BT	CO
1.	Explain the functioning of the Respiratory System (RS).	10	L1	1
2.	Explain lung volumes and capacities with a neatly labeled diagram.	10	L2	2
3.	Describe any four common diseases of the Renal System.	10	L1	3
4.	Explain the structure and functioning of the Renal System.	10	L2	1
5.	Describe common ailments of the respiratory system, their causes, and symptoms.	10	L2	3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	20	10	20	-	20	30	-	-	-	-

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Sl. No.	TEST Part- B Questions	M	
1 a	List the six classical schools of Indian philosophy.	03	
b	Explain any two threats to traditional knowledge.	02	
c	Mention one key difference between traditional Indian and Western systems of knowledge.	02	
d	How does the Vedic Corpus form the foundation of Indian knowledge traditions?	03	
2 a	Draw a neat framework for classification of IKS Corpus	03	
b	Name the six Vedangas.	03	
c	State the two types of perception recognized in Nyaya philosophy	02	
d	Differentiate between agricultural and ecological traditional knowledge.	02	
3 a	Mention any four Yogasutra of Patanjali	03	
b	Describe the characteristics and scope of Indian traditional knowledge in the modern context.	03	
c	Mention one key feature that differentiates Nyaya from Samkhya philosophy.	02	
d	Name the six orthodox (*āstika*) schools of Indian philosophy	02	
4 a	Find out the difference from the Kaṭapayādi system to Bhūta-saṅkhyā for the word "Rāma".	03	
b	Name the three generic measures used in the ancient Indian system.	02	
c	Laghu and Guru syllables in the given word: योगदर्शन (yōgadarśana)	02	
d	Develop a binary cycle of length four and identify all possible binary words of length fours.	03	
5 a	What is the standardized units for street widths in Indus-Saraswati civilization?	02	
b	Who the concept of zero discovered in India? What is its special significance?	03	
c	What is the numeral representation for the word 'manu, agni, elephant, Tithi, rudra, Indra, Pandava.	03	
d	Briefly provide the organization of "Ashtadhyayi". How many rules are there in Ashtadhyayi ?	02	

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	10	10	5	—	15	10	—	—	—	—